

# PWM Servo Introduction

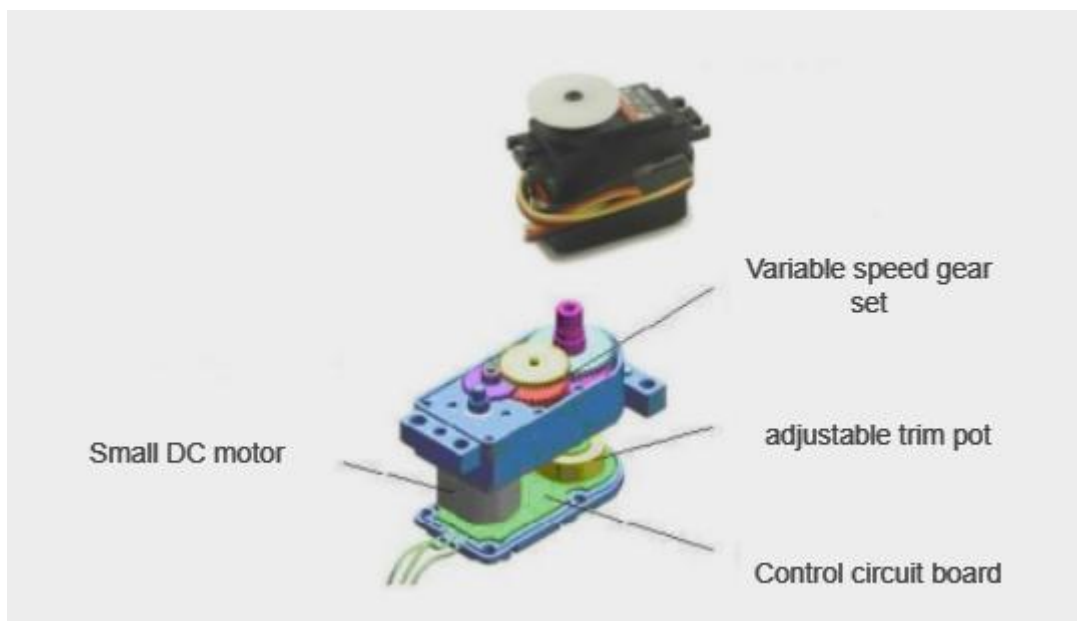
## 1. Servo Working Principle

### 1.1 What is a Servo

A servo, also known as a servo motor, was initially used to control the steering of ships. Due to its ability to precisely control the rotation angle of the rubber, it is commonly used in mechanical systems with high precision requirements. Servos play a crucial role in robots, including both industrial and home service robots, all of which necessitate precise control capabilities. Servos act as the joints in robots, functioning as part of the actuator system. Therefore, servos can be metaphorically described as the muscles of robots, providing them with the necessary power.

### 1.2 How does a Servo Work

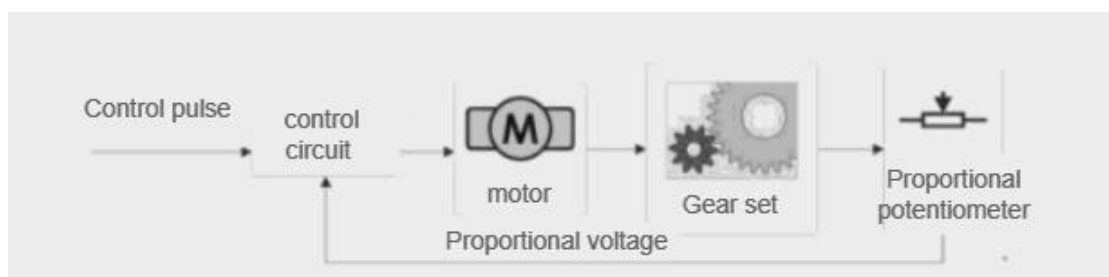
The servo consists of four parts: a motor, a control circuit board, a gear set, and a potentiometer.



A traditional servo has three wires: a power wire, a GND wire, and a signal

control wire.

Its working principle is as follows: when the control circuit board of the servo receives a pulsing signal, it controls the rotation of the motor. The motor drives the output shaft to rotate via a gear set, and the rotation of the output shaft also drives the potentiometer to rotate. The control circuit board detects the rotation of the potentiometer and determines if it has rotated to the specified angle. If it has, the motor stops rotating; otherwise, the motor continues to rotate.



## 1.3 PWM Servo Introduction

A PWM servo is controlled by PWM, which differentiates it from the traditional analog servo that requires continuous PWM signals to maintain a specific position or rotate at a certain speed. However, a PWM servo only needs to receive one PWM signal to remain in a specific position.

Before assembling a servo, it is necessary to adjust the servo to the neutral position.

The neutral position of a servo is its initial position, and it is used as the “zero point” for positive and negative angle rotations. It's necessary to adjust the servo to the neutral position before installing it onto the servo horn.

This is because when a servo rotates, its rotating components drive the potentiometer to rotate, and the software assumes that the neutral position is the “zero point”. Otherwise, the potentiometer may enter a “dead zone”, causing the entire components to malfunction. In robot applications, this can

result in the servo failing to reach the specified angle or producing inconsistent action groups.